Displacement, Velocity and Acceleration

Displacement

$s(t)$: distance and direction an object has moved from an origin over a period of time. (usually in metres, m)

Velocity

$v(t) = s'(t)$: rate of change of displacement with respect to time. (usually in metres per second, m/s)

Acceleration

$a(t) = v'(t) = s''(t)$: rate of change of velocity with respect to time (usually in metres per second squared, m/s^2)

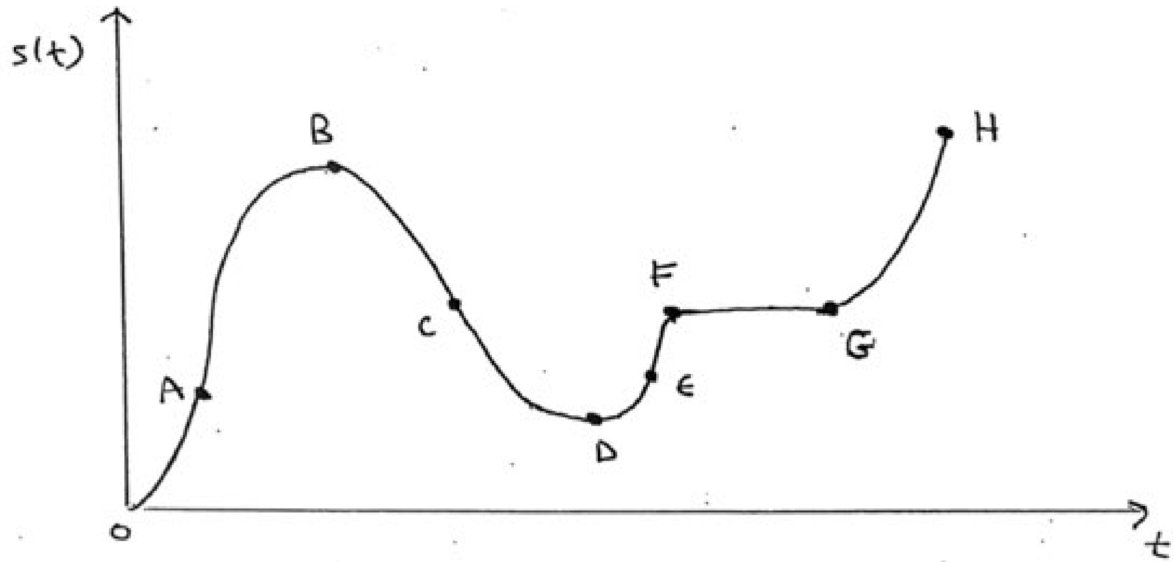
Relating Position, Velocity and Acceleration

1. An object is **speeding up** at time, t if $v(t) \times a(t) > 0$.
2. An object is **slowing down** at time, t if $v(t) \times a(t) < 0$.
3. An object is **moving away** from its original position if $s(t) \times v(t) > 0$.
4. An object is **moving towards** from its original position if $s(t) \times v(t) < 0$

$s(t)$	$v(t)$	$a(t)$	$v(t) \times a(t)$	**Motion of particle** Speeding Up **or** Slowing Down **and** Moving Away **or** Moving Towards	**Description of Slope of** $s(t)$ Positive **or** Negative Slope **and** Increasing **or** Decreasing
+	+	+	+		
-	+	-	-		
+	-	+	-		
-	-	-	+		

Example 1:

The graph of the position function is shown below.



a) Describe the slope of the graph in terms of **positive, negative, increasing,** or **decreasing** over each indicated interval.

b) Determine the sign of the velocity and acceleration.

Interval	Slope of Graph	Velocity	Acceleration
O to A			
A to B			
B to C			
C to D			
D to E			
E to F			
F to G			
G to H			

Position, Velocity, Acceleration

Example 1: The position function of a particle is $s(t) = t^4 - 12t^3 + 30t^2 + 5t,\ t \geq 0$. When is the acceleration positive and negative?

Example 2: A particle moves according to $s(t) = 2t^3 - 21t^2 + 60t - 12,\ t \geq 0$, where s is in metres and t is in seconds.

a. When does the particle change direction?

b. How fast is the particle moving after 3 seconds?

c. Is the particle speeding up or slowing down at 4 seconds?

d. Sketch the path of $s(t)$.

Example 3: On windless day, an archer shoots an arrow vertically upward so that its height in metres above the ground t seconds after release is given by the formula $h(t) = -4.9t^2 + 24.5t + 2$.

a. Determine the velocity at any time.

b. When is the velocity at -5 m/s?

c. When does the arrow hit the ground?

d. At what time does the arrow stop rising?